

Lesson 15 — Design Challenge — Logic Face-Off

Matches LEGO Teacher Portal lesson: "Smart Park".

Objective (I can...): I can design an autonomous machine that makes layered decisions, and compare my logic with another team's. **CS concept:** Applying selection, compound conditions, and thresholds to an autonomous build · **Builds on:** Lessons 11-14. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): A self-driving system layers decisions every moment — obstacle? pedestrian? light color? *First-principles question:* "What sensor + rule would make YOUR machine make a smart, defensible decision?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, trigger objects, notebooks. **Model already built** (Conditionals model) — the 30 min is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Logic review — pairs match scenarios to if/then, if/then/else, compound, and multi-way. (*Theme: Logic face-off.*) - **Concept + demo (10):** Challenge: build an autonomous "guard," "sorter," or "navigator" that decides among actions using at least a compound condition or a threshold. Coach models a flowchart for the decision, then hands off. - **Build/Code (30):** Pairs plan (flowchart) and code a layered decision on the built model. *Swap at 15 min.* - **Challenge (7):** Add a second decision or wrap the decision in a loop so it decides continuously. - **Share + pack-up (5):** **Logic face-off** — pair two teams that met the *same* goal; each traces their logic for the same test input, then together they judge which design is clearer and why (often more than one correct answer).

Photograph each build.

Pair roles: Co-plan; Driver codes / Navigator checks the logic and flowchart; swap at 15 min. (*Trio rotates plan/code/trace.*)

Differentiation: *Support* — a single-condition target with a partial flowchart. *Extension / fast-finisher* — require a compound condition inside a loop. *Reteach* — reuse a Lesson 13-14 program as a template.

Exit check: Pair delivers an autonomous model that decides correctly with layered logic and can trace it against another team's solution. **Progress layer:** Send home the **Unit 3 parent note** ("your student can program a machine to make its own layered decisions") with the **build photo**. **At-home (optional):** Design a "smart" rule-set for a gadget using AND/OR and a threshold; write it as a flowchart. ~10 min.